

12 DRILL: the practice set with answers

Solve, then check. Each item names its distribution and slot. Answers are exact where the source gives an exact value.

The drill as a progress ladder

RUNG	1	fundamentals	_____	D1-D3	probability, counting, complements
RUNG	2	discrete	dists _____	D4-D17	binomial, Poisson
RUNG	3	continuous	dists _____	D18-D31	uniform, normal, exponential
RUNG	4	Chebyshev	_____	D32-D36	bound, complement, inverse
RUNG	5	sampling	/CLT _____	D37-D41	SE, z, the CLT table
RUNG	6	estimation	_____	D42-D46	compare, point, interval
RUNG	7	siblings	_____	D47-D50	the mutation surface

R7	siblings (the risk surface)
R6	estimation
R5	sampling / CLT
R4	Chebyshev
R3	continuous distributions
R2	discrete distributions
R1	fundamentals

climb in order. a rung is "done" when you can do it without notes.

Warm-up: probability and counting

D1	Six faculty, choose 3 for laptops. How many ways?	
ans	$C(6,3) = 20$	
D2	A camera chain: P(works each step) given. P(all three work)?	
ans	product of the three, from the multiplication rule	
D3	"At least one" heads in 3 fair coins.	
ans	$1 - (1/2)^3 = 7/8$	(complement)

Binomial

D4	B(12, 0.1). $P(X=2)$.	ans $66 \times 0.01 \times 0.348678 = 0.2301$
D5	B(12, 0.1). $P(X \geq 2)$.	ans $1 - 0.2824 - 0.3766 = 0.3410$
D6	B(12, 0.1). $P(X=0)$.	ans 0.2824
D7	Irregular die, $P(X=5)=2P(X=4)$. Find p.	ans 5/8
D8	Same die, 10 throws, P(no even).	ans $(3/8)^{10} = 5.499e-5$
D9	Out of 10000 sets of 10, expected with no even number.	ans 0.5499
D10	B(600, 1/6). E and Var.	ans 100, 250/3

Poisson

D1 1	$P(X=1)=0.2$ $P(X=2)$. Find λ . ans $\lambda = 10$	
D1 2	Same, $P(X=0)$.	ans $e^{-10} = 4.54e-5$
D1 3	Calls at 2/min. $P(\text{exactly } 2 \text{ in a minute})$.	ans $2e^{-2} = 0.2707$
D1 4	$P(\text{exactly } 2 \text{ in each of } 5 \text{ minutes})$.	ans $(2e^{-2})^5 = 32e^{-10} = 0.00145$
D1 5	5000 men, death rate 0.001. $P(X=4)$.	ans $e^{-5} 5^4/4! = 0.1755$
D1 6	$P(X>2)$ with $\lambda=5$.	ans 0.8753
D1 7	"Rejected" $P(X\geq 4)$ with $\lambda=2.5$. ans 0.2424	

Uniform

D18	$U(2,6)$. f, mean, variance.	ans 0.25, 4, 1.3333
D19	$U(2,6)$. $P(3<X<5)$.	ans $2/4 = 0.50$
D20	$U(2,6)$. $P(X<1)$.	ans 0 (below support)

Normal

D21	$\mu=8$, $sd=5$. $P(X<8.6)$.	ans $Z=0.12$, 0.5478
D22	$\mu=8$, $sd=5$. $P(X>8.6)$.	ans $1 - 0.5478 = 0.4522$
D23	$\mu=8$, $sd=5$. Find x for $P(X<x)=0.20$. ans $Z=-0.8416$, $x = 8 - 4.208 = 3.792$	
D24	Marks: pass 40 = 46 percent, distinction 75 = 9 percent. Find μ , σ .	
	ans $\sigma = 28.2$, $\mu = 37.2$	
D25	Same, re-exam cutoff at the 24 percent lower tail.	ans 30.4
D26	$N(100, 50^2)$, $X=200$. Find Z .	ans 2.0

Exponential and related continuous models

D27	15/hour. $P(\text{gap} < 3 \text{ min})$.	ans 0.05 hr, $1-e^{-0.75} = 0.5276$
D28	15/hour. Mean gap.	ans 4 minutes
D29	Mean 2. $P(X<1 \mid X<2)$.	ans 0.6225
D30	Gamma-shaped pdf $f(x)=x e^{-(x/3)}/9$, $P(X>12)$. ans $5e^{-4} = 0.0916$	
This is not an exponential density, even though its tail calculation uses e.		
D31	Memoryless: $P(T>8 \mid T>5)$.	ans $P(T>3) = e^{-3\lambda}$

Chebyshev

D3 2	Two dice, $P(X-7 \geq 3)$ bound.	ans $\sigma^2=35/6$, $k=3$, $35/54 = 0.6481$, exact $1/3$
D3 3	600 throws, $P(80 \leq X \leq 120)$ lower bound.	ans $19/24 = 0.7917$
D3 4	Mean 70, var 25, $P(60 < X < 80)$ lower bound.	ans $k=2$, 0.75
D3 5	$\mu=10$, $\text{var}=4$, $P(5 < X < 15)$ and $P(X-10 \geq 3)$.	ans $21/25=0.84$, and $4/9=0.4444$
D3 6	Find c so $P(X-10 \geq c) \leq 0.04$, var 4.	ans $k=5$, $c=10$

Sampling, CLT

D37	$\sigma=100$, $n=25$ then $n=100$. Find SE.	ans 20 then 10
D38	$\mu=4$, $\sigma=2$, $n=36$. $P(\bar{X} > 4.5)$.	ans $Z=1.50$, 0.0668
D39	$\mu=4.0$, $\sigma=1.5$, $n=50$. $P(3.5 < \bar{X} < 3.8)$.	
	ans $Z1=-2.36$, $Z2=-0.94$; 0.1645 from rounded table areas, about 0.1637 unrounded	
D40	$\mu=50000$, $\sigma=8000$, $n=64$. $P(\bar{X} < 48000)$. ans $Z=-2.00$, 0.0228	
D41	$\mu=7$, $\sigma=1$, $n=9$ (normal pop). $P(6.4 < \bar{X} < 7.2)$.	ans $Z1=-1.80$, $Z2=0.60$, 0.6898

Estimation

D42	Sample of 5 normal; $t1=\text{mean}$, $t2=(X1+X2)/2+X3$, $t3=(2X1+X2+\lambda X3)/3$.	
	Find λ , the unbiased ones, variances, the best.	
	ans $\lambda=0$; $t1$ and $t3$ unbiased, $t2$ not; Var 0.2, 1.5, 0.5556 σ^2 ; best $t1$	
D43	$T1=X1+X2-X3$, $T2=2X1+3X3-4X2$, $T3=(\lambda X1+X2+X3)/3$.	
	Find λ , unbiased set, variances, best, and comment on consistency.	
	ans $\lambda=1$; all unbiased; Var 3, 29, $1/3 \sigma^2$; best $T3$. Fixed $n=3$ does not	
	prove consistency; the sequence $\bar{X}_n$ is consistent under usual iid assumptions.	
D44	OUT OF MTE SCOPE: $n=25$, $\bar{X}=8.4$, $\sigma=1.5$, 95 percent interval.	
	ans $8.4 \pm 0.588 = (7.81, 8.99)$	
D45	Response times sum 1640, $n=8$. Point estimate.	ans 205 ms
D46	465 of 500 packets succeed. Point estimate of p .	ans 0.93

The four high-risk siblings (from 07-PATTERNS, do these)

D47	Normal one-unknown: 30 percent below 45 and 8 percent above 64, find μ , σ .
D48	Binomial two-sided interval: $B(12, 0.1)$, $P(2 \leq X \leq 4)$?
D49	CLT for a SUM: $n=40$ draws, $\mu=10$, $\sigma=3$. $P(\text{sum} > 440)$?
ans sum $\sim N(400, 360)$, $Z = (440-400)/\sqrt{360} = 2.108$, tail 0.0175	
D50	Discrete conditional: $B(10, 0.5)$, $P(X=5 \mid X \geq 3) = P(5)/(1-P(0)-P(1)-P(2))$

The self-check rule

if you can do D4-D46 without notes:
 Section B and C of the paper are MECHANICAL
 if D47-D50 are also fine:
 you are covered on the MUTATION SURFACE
 if anything above is shaky:
 go back to 04-METHODS.md for that SLOT before more drills